A LOCAL RING IS UNIBRANCH EXACTLY WHEN ITS HENSELIZATION HAS A UNIQUE MINIMAL PRIME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, together with the project's own definitions of the reference/slogan/history display environments, which the project itself typesets as comments; nothing was added, removed or redefined by this card and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span includes the author's own \begin{reference} block citing the published source of the equivalence; it is delivered verbatim and produces no unresolved citation. The author writes that the result 'follows from' the branch lemma but gives a direct proof, which is what the counted proof span contains. The eight references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Henselization of a local ring, its flatness, and the identity (A/p)^h = A^h/pA^h. used at proof lines 31269-31276. Is what identifies the unique minimal prime of A^h with the extension of that of A.
- Unibranchness as locality of the integral closure of the reduced quotient. used at proof lines 31277-31278. Fixes the equivalent form of the conclusion that the direct proof establishes.
- Integrality, going up, and flat base change of a chain of ring inclusions. used at proof lines 31279-31301. Produces the chain of inclusions over A^h whose maximal ideals give the contradiction.

Source reference. Original lemma 00HS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 654–654. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 092Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11496–11496. Cross-chapter navigation pointer into

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04GJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5835–5835. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 033C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4051–4051. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7594–7594. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 31155: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31155–31155. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 11067: sources/stacks-more-algebra/pinned-original-source.tex, source lines 11067–11067. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a local ring. Let A^h be the henselization of A. The following are equivalent*

(1) *A is unibranch, and*
(2) *A^h has a unique minimal prime.*

Proof. This follows from Lemma 31155 but we will also give a direct proof. Denote $\mathfrak{m}$ the maximal ideal of the ring A. Recall that the residue field $\kappa = A/\mathfrak{m}$ is the same as the residue field of A^h.

Assume (2). Let $\mathfrak{p}^h$ be the unique minimal prime of A^h. The flatness of $A \to A^h$ implies that $\mathfrak{p} = A \cap \mathfrak{p}^h$ is the unique minimal prime of A (by going down, see Algebra, Lemma 00HS). Also, since $A^h/\mathfrak{p}A^h = (A/\mathfrak{p})^h$ (see Algebra, Lemma 05WQ) is reduced by Lemma 11067 we see that $\mathfrak{p}^h = \mathfrak{p}A^h$. Let A' be the integral closure of $A/\mathfrak{p}$ in its fraction field. We have to show that A' is local. Since $A \to A'$ is integral, every maximal ideal of A' lies over $\mathfrak{m}$ (by going up for integral ring maps, see Algebra, Lemma 00GU). If A' is not local, then we can find distinct maximal ideals $\mathfrak{m}_1$, $\mathfrak{m}_2$. Choose elements $f_1, f_2 \in A'$ with $f_i \in \mathfrak{m}_i$ and $f_i \notin \mathfrak{m}_{3-i}$. We find a finite subalgebra $B = A/\mathfrak{p}[f_1, f_2] \subset A'$ with distinct maximal ideals $B \cap \mathfrak{m}_i$, $i = 1, 2$. Note that the inclusions

$$A/\mathfrak{p} \subset B \subset \kappa(\mathfrak{p})$$

give, on tensoring with the flat ring map $A \to A^h$ the inclusions

$$A^h/\mathfrak{p}^h \subset B \otimes_A A^h \subset \kappa(\mathfrak{p}) \otimes_A A^h \subset \kappa(\mathfrak{p}^h)$$

the last inclusion because $\kappa(\mathfrak{p}) \otimes_A A^h = \kappa(\mathfrak{p}) \otimes_{A/\mathfrak{p}} A^h/\mathfrak{p}^h$ is a localization of the domain $A^h/\mathfrak{p}^h$. Note that $B \otimes_A \kappa$ has at least two maximal ideals because $B/\mathfrak{m}B$ has two maximal ideals. Hence, as A^h is henselian we see that $B \otimes_A A^h$ is a product of ≥ 2 local rings, see Algebra, Lemma 04GJ. But we've just seen that $B \otimes_A A^h$ is a subring of a domain and we get a contradiction.

Assume (1). Let $\mathfrak{p} \subset A$ be the unique minimal prime and let A' be the integral closure of $A/\mathfrak{p}$ in its fraction field. Let $A \to B$ be a local map of local rings inducing an isomorphism of residue fields which is a localization of an étale A-algebra. In particular $\mathfrak{m}_B$ is the unique prime containing $\mathfrak{m}B$. Then $B' = A' \otimes_A B$ is integral over B and the assumption that $A \to A'$ is local implies that B' is local (Algebra, Lemma 092Y). On the other hand, $A' \to B'$ is the localization of an étale ring map, hence B' is normal, see Algebra, Lemma 033C. Thus B' is a (local) normal domain. Finally, we have

$$B/\mathfrak{p}B \subset B \otimes_A \kappa(\mathfrak{p}) = B' \otimes_{A'} (\text{fraction field of } A') \subset \text{fraction field of } B'$$

Hence $B/\mathfrak{p}B$ is a domain, which implies that B has a unique minimal prime (since by flatness of $A \to B$ these all have to lie over $\mathfrak{p}$). Since A^h is a filtered colimit of the local rings B it follows that A^h has a unique minimal prime. Namely, if $fg = 0$ in A^h for some non-nilpotent elements f, g, then we can find a B as above containing both f and g which leads to a contradiction. □